EVS NOTES

SYLLABUS:

The Multidisciplinary Nature of Environmental Studies Definition, scope and importance of Environmental Studies. Biotic and a biotic component of environment, need for environmental awareness.

Ecosystems: Concept of an ecosystem, structure and function of an ecosystem, producers, consumers and decomposers, energy flow in the ecosystem, ecological succession, food chains, food webs and ecological pyramids. Introduction, types, characteristic features, structures and function of different ecosystem.

Bio-diversity and its Conservation: Introduction to biodiversity — definition: genetic, species and ecosystem diversity, Bio-geographical classification of India, Value of biodiversity: Consumptive use, productive use, social, ethical, aesthetic and option values, Biodiversity at global, national and local levels, India as a mega-diversity nation, Hot-spots of biodiversity, Threats to biodiversity: Habitat loss, Poaching of wildlife, man–wildlife conflicts, rare endangered and threatened species (RET), endemic species of India, method of biodiversity conservation: In-situ and ex-situ conservation.

The Multidisciplinary Nature of Environmental Studies

Environmental studies is a broad academic field that systematically studies the complex interactions between humans and the natural environment. To effectively understand and address the intricate challenges facing our planet, this discipline must integrate knowledge from a wide range of fields. It is a synthesis of the natural sciences, social sciences, and humanities, all working together to find sustainable solutions.

The Multidisciplinary Nature of Environmental Studies

1. Biological Sciences 🌱

- Deals with living organisms: plants, animals, microorganisms.

- Helps in understanding **ecosystems, food chains, biodiversity, and conservation**.
- Explains how human activities affect flora & fauna.

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2. Physical Sciences ⚛️

- **Physics** explains energy, radiation, climate systems.
- **Chemistry** explains pollutants, air/water quality, and biogeochemical cycles.
- Provides basis for pollution control and renewable energy.

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3. Earth Sciences 🌍

- Includes geology, soil science, hydrology, and atmospheric studies.
- Helps in understanding **natural resources, disasters (earthquakes, floods)**.
- Explains soil erosion, land use, and water cycle management.

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4. Social Sciences 👥

- **Economics** studies use of resources, cost-benefit of conservation.
- **Sociology** studies population growth, cultural practices, and awareness.
- **Political science** explains policies and government initiatives.

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5. Medical Sciences 🏥

- Studies the **impact of environment on human health**.
- Explains pollution-related diseases, sanitation, and safe drinking water.
- Guides policies for environmental health and hygiene.

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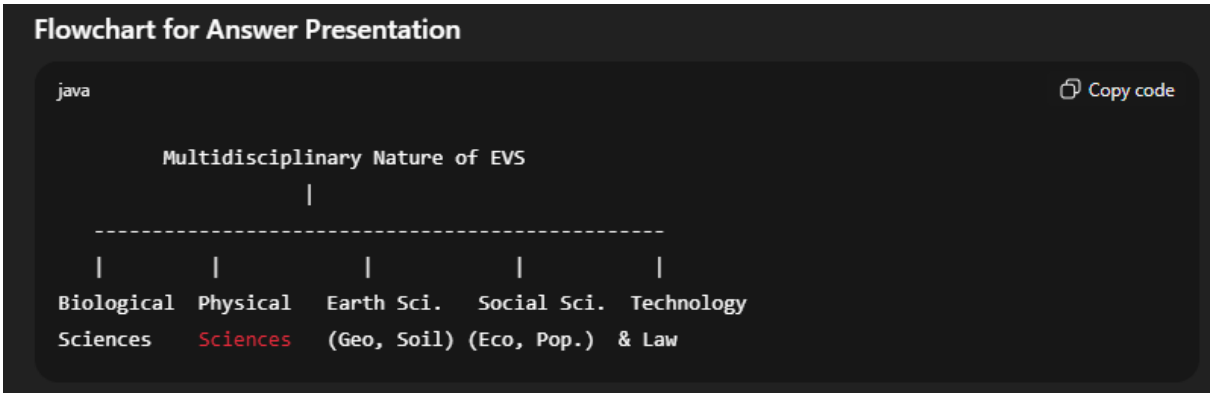
6. Engineering & Technology ⚙️

- Develops eco-friendly technologies like **solar, wind, hydro power**.
- Designs waste management, recycling, sewage treatment.
- Provides technical solutions to minimize pollution.

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7. Law and Management ⚖️

- **Environmental laws** protect forests, wildlife, and reduce pollution.
- **Management principles** ensure sustainable use of resources.
- Helps in Environmental Impact Assessment (EIA) and planning.



Definition, Scope and Importance of Environmental Studies

1. Definition

- Environmental Studies is a **multidisciplinary field** that deals with the interaction of humans with the environment.
- It focuses on the study of **natural resources, ecology, biodiversity, pollution, social issues, and sustainable development**.

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2. Scope of Environmental Studies

The scope covers **various areas**:

1. **Natural Resources** – Study of renewable and non-renewable resources and their sustainable use.
2. **Ecosystem and Biodiversity** – Understanding ecological balance and conservation of biodiversity.
3. **Pollution Control** – Identifying causes, effects, and solutions of air, water, soil, and noise pollution.

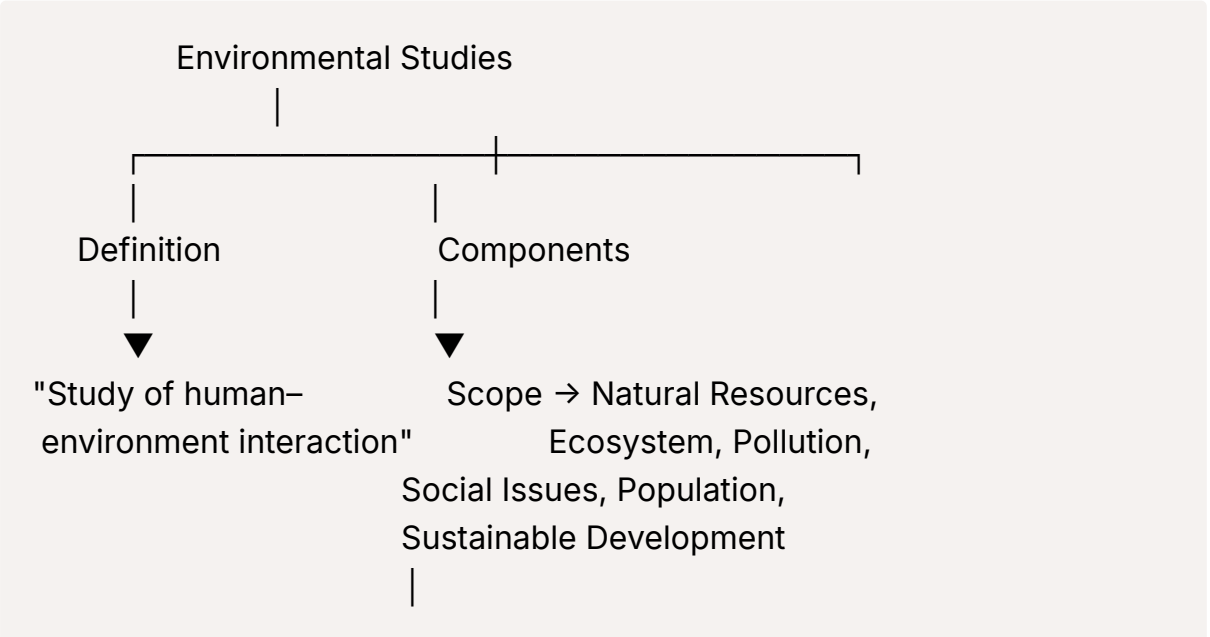
4. **Social Issues and Environment** – Environmental ethics, global warming, climate change, deforestation.
5. **Human Population and Environment** – Impact of population growth, urbanization, and industrialization.
6. **Sustainable Development** – Balancing human needs with environmental protection for future generations.

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3. Importance of Environmental Studies

1. **Awareness** – Creates awareness about environmental problems and solutions.
2. **Conservation** – Promotes conservation of forests, wildlife, and natural resources.
3. **Pollution Control** – Helps in finding ways to reduce environmental pollution.
4. **Better Quality of Life** – Ensures healthy living conditions by protecting air, water, and soil.
5. **Sustainable Future** – Encourages sustainable practices for long-term ecological balance.
6. **Decision Making** – Guides government and society in making eco-friendly policies.

Flowchart (Exam Presentation)



▼

Importance → Awareness,
Conservation,
Pollution Control,
Better Life,
Sustainable Future

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1. Biotic Components (Living factors)

These are all living organisms in an ecosystem. They are grouped by their role.

A. Producers (Autotrophs)

- **What:** Green plants, algae and some bacteria that make organic food from inorganic materials (CO_2 + water) using sunlight (photosynthesis).
- **Why important:** They form the **foundation of energy flow** in ecosystems — all other organisms depend (directly/indirectly) on them.
- **Examples & notes:** Trees, grasses, phytoplankton. High productivity (e.g., tropical forests, algal blooms) → more biomass and higher food-chain support.
- **Exam tip:** Mention “primary productivity” (rate of biomass production) when asked.

B. Consumers (Heterotrophs)

- **Primary consumers (Herbivores):** Eat producers (e.g., deer, insects). Transfer energy from plants to higher levels.
- **Secondary consumers (Carnivores/Omnivores):** Eat herbivores (e.g., frogs, small snakes); omnivores eat both (e.g., humans, bears).
- **Tertiary/top predators:** Feed on other carnivores (e.g., lion, eagle); control population of lower levels.
- **Role & effect:** Consumers regulate population sizes, influence species composition, and move energy up trophic levels.
- **Example point:** Overhunting top predators can cause trophic cascades (increase in herbivores → overgrazing).

C. Decomposers & Detritivores

- **What:** Bacteria, fungi (decomposers) and organisms like earthworms, termites (detritivores) that break down dead organic matter.
- **Function:** Recycle nutrients (N, P, K…) back into soil and water — essential for plant growth and ecosystem recycling.
- **Importance:** Without decomposers, nutrients would remain locked in dead matter and primary productivity would fall.
- **Example:** Leaf litter → fungi/bacteria → mineral nutrients → taken up by plants.

D. Biotic Interactions (short)

- **Predation, competition, mutualism, parasitism, commensalism.** These interactions shape community structure and species distribution.
- **Food chains & webs:** Linear feeding sequences (food chain) combine into complex networks (food web); energy decreases at each trophic level (10% rule).

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2. Abiotic Components (Non-living factors)

These are physical and chemical factors that determine habitat conditions and limit which organisms can live where.

A. Climatic Factors

1. **Sunlight (Light)**
    - Controls photosynthesis, plant growth, and seasonal behavior (flowering, migration).
    - Intensity and photoperiod (day length) affect primary productivity and life cycles (e.g., flowering, breeding).
2. **Temperature**
    - Affects metabolic rates, enzyme activity and species' geographic distribution.

- Extremes cause heat stress or freeze damage; many organisms have thermal tolerance ranges.

3. **Precipitation (Rainfall) & Water Availability**
    - Determines habitat types (desert, grassland, forest) and soil moisture.
    - Influences plant growth, groundwater recharge and water-dependent life stages (e.g., amphibian breeding).
4. **Humidity**
    - Affects transpiration in plants, evaporative cooling in animals, and microclimate conditions.
5. **Wind**
    - Influences temperature, transpiration, seed/pollen dispersal and erosion.
    - Strong winds shape plant form (stunted trees, wind-pruned shapes).

Why climate matters: Climatic patterns (rainfall + temperature) largely determine ecosystem type and species composition.

B. Edaphic (Soil) Factors

1. **Soil Texture & Structure**
    - Texture (sand/silt/clay) affects water retention and aeration; structure affects root penetration and drainage.
    - Sandy soils drain fast (low water retention); clay holds water but may be poorly aerated.
2. **Soil pH**
    - Affects nutrient availability and microbial activity.
    - Acidic or alkaline soils limit certain plant species; neutral soils often most fertile.
3. **Organic Matter & Fertility**
    - Humus provides nutrients and improves soil structure; more organic matter → higher fertility.
    - Decomposer activity returns organic nutrients to the soil.

4. **Soil Moisture & Drainage**
    - Determines oxygen availability to roots and soil organisms; waterlogged soil → anaerobic conditions harmful to many plants.
5. **Mineral Content & Salinity**
    - Levels of N, P, K, Ca, Mg, and trace elements determine plant nutrition.
    - High salinity reduces plant growth and can lead to barren soils.

Why soil matters: Soil is the medium for plant growth and hosts many organisms; it controls productivity and vegetation types.

C. Topographic Factors

1. **Altitude (Elevation)**
    - Temperature and oxygen decrease with altitude → alpine vegetation and adapted species.
    - Example: Forest → shrub → grass → alpine depending on elevation.
2. **Slope & Aspect (Direction of slope)**
    - Slope affects drainage and soil depth; steep slopes have thinner soil and more runoff.
    - Aspect influences sunlight exposure — south-facing slopes (in Northern Hemisphere) get more sun → warmer/drier microclimates.
3. **Relief & Landforms**
    - Mountains, valleys, plains create varied microclimates and habitats; influence rainfall patterns (orographic rainfall).

Why topography matters: Creates microhabitats and spatial variation in ecosystems.

D. Chemical Factors (Water/Atmospheric chemistry & nutrients)

1. **Water Chemistry**
    - Dissolved oxygen, pH, salinity in aquatic systems determine which organisms can live there (fish, plankton).
    - Pollutants (heavy metals, nitrates) alter water quality.

2. **Gases: Oxygen & Carbon Dioxide**
    - Oxygen is essential for aerobic respiration; CO_2 is required for photosynthesis.
    - Changes (e.g., hypoxia in water bodies) can cause mass die-offs.
3. **Nutrients (N, P, K, micronutrients)**
    - Determine plant growth; nutrient limitation limits primary productivity.
    - Excess nutrients (eutrophication) can cause algal blooms and oxygen depletion.
4. **pH & Chemical Pollutants**
    - Strongly acidic or alkaline conditions hamper life; pollutants (acid rain, toxic wastes) can disrupt ecosystems.

Why chemicals matter: Provide materials for life processes but become harmful when out of balance.

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3. How Abiotic and Biotic Factors Interact (short)

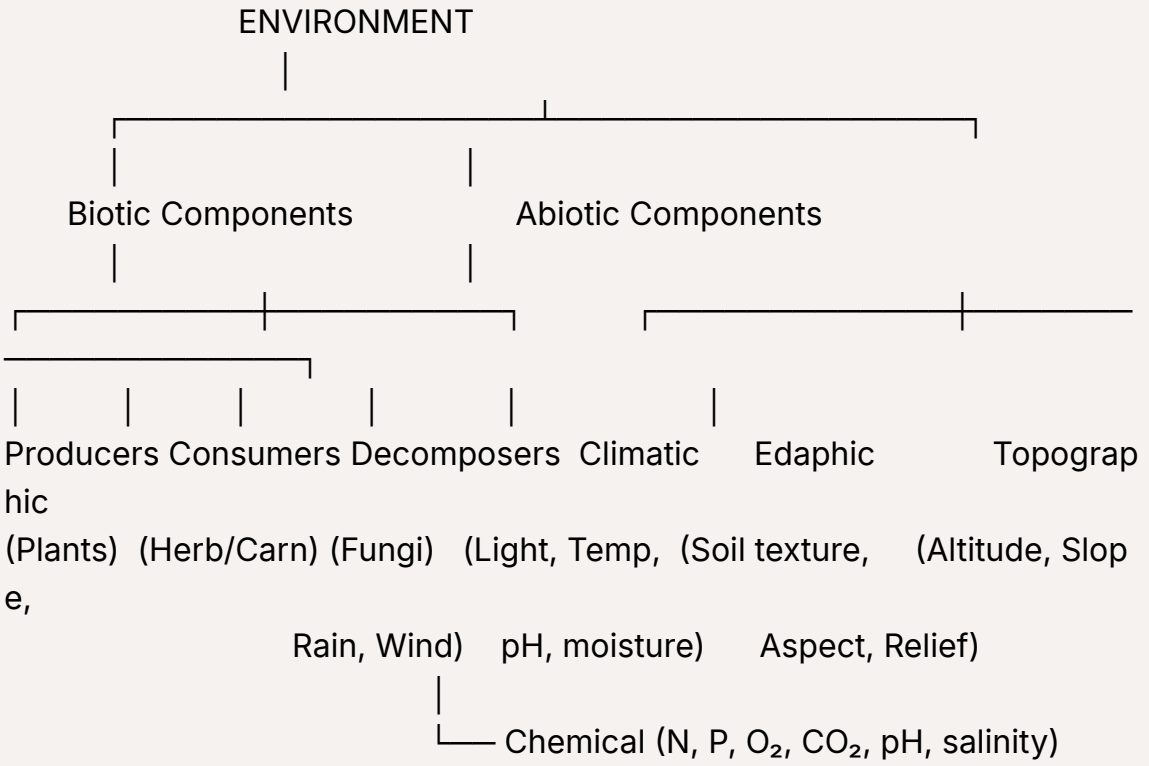
- **Abiotic sets the stage; biotic fills it.** Climate + soil + topography determine which producers grow → consumers and decomposers follow.
- **Feedback loop:** Biotic processes (plant cover, decomposition) modify abiotic conditions (soil organic matter, microclimate).
- **Example:** Forest vegetation (biotic) increases soil humus and water retention (abiotic), which supports more plants.

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4. Quick Examples (use in answers)

- **Desert:** Low rainfall (abiotic) → sparse xerophytic plants (biotic), animals adapted to conserve water.
- **Mangrove:** High salinity & tidal inundation (abiotic) → salt-tolerant trees (biotic) that protect shorelines.
- **Alpine zone:** Low temp & high UV (abiotic) → dwarf plants and specialized animals (biotic).

Flowchart (copy into exam — neat boxes + arrows)



Exam-writing tips (2 lines)

- Start with **one-line intro**, then list each factor as a heading + 2 short lines (definition + role/example).
- Draw the **flowchart** and add one line on **interactions/importance** as conclusion.

FROM HERE JUST GIVING KEY POINTS U CAN MAKE SENTENCES BASED ON WORDS (EK RAT PEHKE VESE BHI YAD

NHI HOTA AND EVS YAD KARNE KA SUBJECT NAHI HAI)

Need for Environmental Awareness

1. Introduction

- Environmental awareness means **understanding the importance of protecting our environment and using natural resources wisely**.
- It is essential for promoting **sustainable development** and ensuring a **healthy future** for all living beings.

2. Need for Environmental Awareness (Explained in Points)

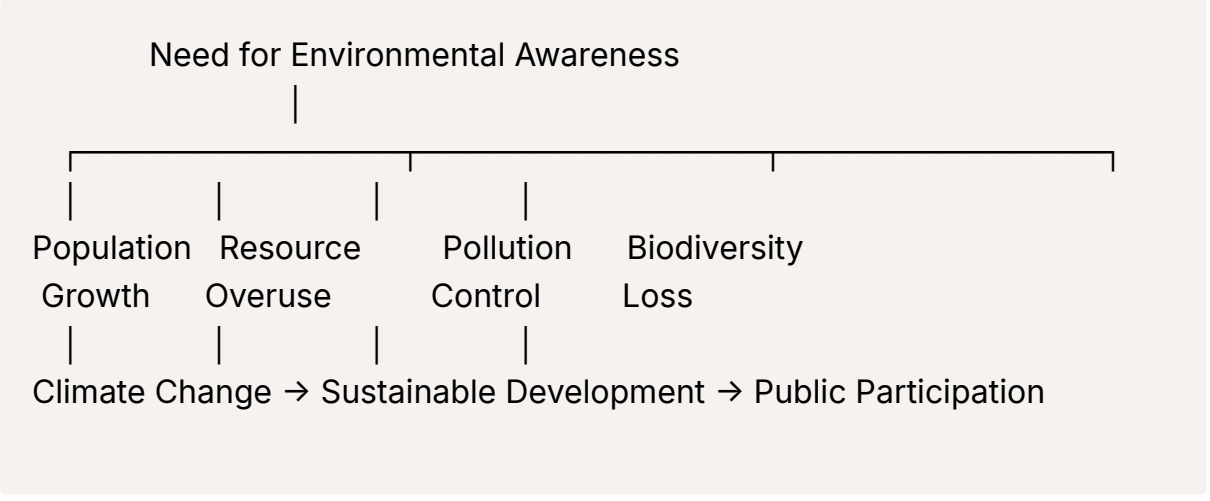
1. **Rapid Population Growth** 👥
   - Increasing population puts pressure on land, water, forests, and energy.
   - Awareness helps people adopt sustainable lifestyles.
2. **Over-Exploitation of Resources** 🌍
   - Excessive use of forests, water, minerals, and fossil fuels is leading to scarcity.
   - Awareness encourages conservation and efficient use of resources.
3. **Environmental Pollution** 🏭
   - Air, water, soil, and noise pollution are causing serious health hazards.
   - Awareness is needed to adopt eco-friendly practices and reduce pollution.
4. **Loss of Biodiversity** 🐅🌱
   - Deforestation, habitat destruction, and hunting are endangering species.
   - Awareness promotes protection of wildlife and ecosystems.
5. **Climate Change & Global Warming** 🌡️

- Rising temperatures, melting glaciers, and extreme weather threaten life.
- Awareness motivates people to reduce carbon footprint and adopt renewable energy.

6. **Need for Sustainable Development ♻️**
   - Present development should not compromise future generations' needs.
   - Awareness teaches balance between **economic growth and environmental protection**.
7. **Public Participation in Policies 🏛️**
   - Environmental laws and policies succeed only when people are aware.
   - Awareness leads to active involvement in movements and decision-making.

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3. Flowchart (Exam Presentation)



(In exam: draw boxes neatly, keep arrows flowing downward or sideways for clarity.)

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4. Conclusion

- Environmental awareness is **not optional but necessary** for survival.
- It ensures **resource conservation, pollution control, biodiversity protection, and sustainable living**.
- Aware citizens can create a **green, safe, and balanced future**.

Concept of an Ecosystem

1. Introduction

- An **ecosystem** is a functional unit of nature where **living organisms (biotic)** interact with **non-living components (abiotic)** in a specific area.
- It involves the **flow of energy** and the **cycling of nutrients**, which maintains balance in nature.

2. Components of an Ecosystem

1. **Biotic Components (Living factors)**
    - **Producers**: Green plants, algae – prepare food via photosynthesis.
    - **Consumers**: Herbivores, carnivores, omnivores – depend on producers or other organisms for food.
    - **Decomposers**: Fungi, bacteria – break down dead matter and recycle nutrients.
2. **Abiotic Components (Non-living factors)**
    - Physical: Sunlight, temperature, rainfall, soil, wind.
    - Chemical: Nutrients, minerals, gases (O_2, CO_2), water pH, salinity.

3. Functions of an Ecosystem

1. **Energy Flow** ⚡
    - Energy enters through sunlight → captured by producers → transferred to consumers → lost as heat.
2. **Nutrient Cycling** ♻️
    - Decomposers recycle essential nutrients (N, P, C) back into soil/water.
3. **Food Chain & Food Web** 🌱🐇🦅
    - Organisms are linked by feeding relationships.
    - Multiple food chains form complex food webs, ensuring stability.
4. **Ecological Balance** ⚖️

- Interaction of biotic and abiotic components maintains balance of population, resources, and environment.

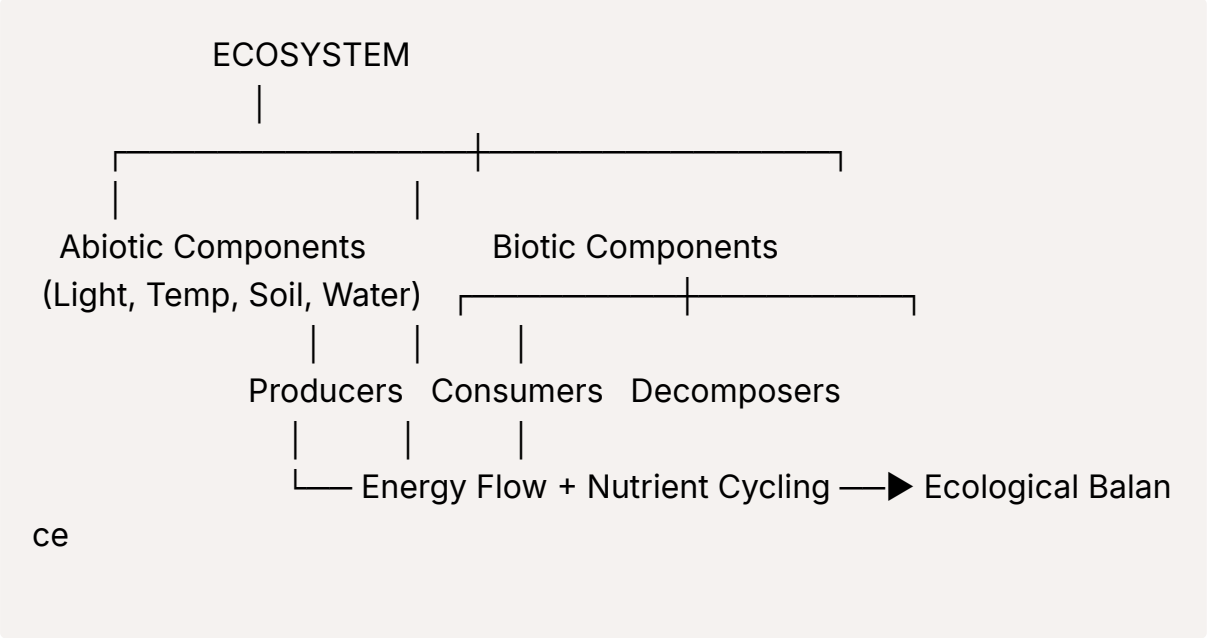
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4. Types of Ecosystems (Brief)

- **Natural**: Forests, deserts, ponds, oceans, grasslands.
- **Artificial/Man-made**: Agricultural fields, aquariums, urban parks.

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5. Flowchart (Exam Presentation)



(Draw with neat boxes & arrows in your answer sheet – adds presentation marks.)

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6. Conclusion

- An ecosystem is a **self-sustaining functional unit** of nature.
- It maintains **life support, energy flow, nutrient cycling, and ecological balance**.
- Understanding ecosystems is essential for **conservation and sustainable development**.

Structure and Function of an Ecosystem

Introduction

- An ecosystem is the **basic functional unit of ecology**, where living (biotic) and non-living (abiotic) components interact.
- It includes **plants, animals, microorganisms, soil, water, air, and climate**.
- The main functions are **energy flow, material cycling, and maintaining ecological balance**.

Structure of an Ecosystem

1. **Abiotic Components** (Non-living factors)
    - **Physical factors** → sunlight, temperature, wind, soil, water.
    - **Chemical factors** → carbon, nitrogen, phosphorus, oxygen, minerals.
    - They provide the **base environment** for life.
2. **Biotic Components** (Living factors)
    - **Producers (Autotrophs)** → Plants, algae, some bacteria. Prepare food by photosynthesis.
    - **Consumers (Heterotrophs)** → Depend on producers for food.
        - Primary consumers → herbivores (deer, grasshopper).
        - Secondary consumers → carnivores (frog, wolf).
        - Tertiary consumers → top predators (lion, tiger).
    - **Decomposers (Saprotrophs)** → Bacteria, fungi, earthworms. Break down organic matter → recycle nutrients.

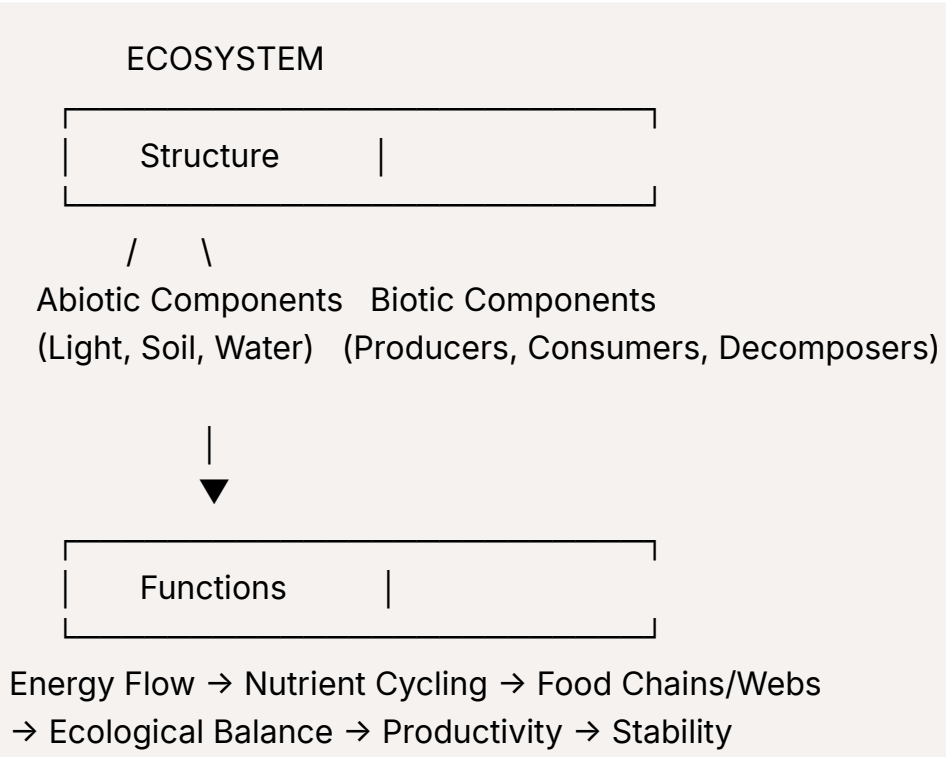
Functions of an Ecosystem

1. **Energy Flow**
    - Energy from the Sun flows **unidirectionally** from producers → consumers → decomposers.
    - Follows **10% energy law** (only 10% of energy passes to next level).
2. **Nutrient Cycling (Biogeochemical cycles)**
    - Elements like carbon, nitrogen, oxygen, phosphorus circulate between living and non-living components.

- Maintains ecosystem productivity.

3. **Food Chains and Food Webs**
   - Food chain = Linear feeding relationship.
   - Food web = Complex interlinked food chains for stability.
4. **Ecological Balance**
   - Keeps populations regulated (no overgrowth of species).
   - Prevents depletion of resources.
5. **Productivity**
   - **Primary productivity** → Energy captured by producers.
   - **Secondary productivity** → Energy stored by consumers.
6. **Stability and Sustainability**
   - Self-regulating system.
   - Can adjust to small disturbances naturally.

Flowchart: Structure and Function of Ecosystem



Conclusion

- An ecosystem is a **self-sustaining functional unit** of nature.
- Its structure includes **biotic and abiotic components**, while its function includes **energy flow, cycling of nutrients, and maintaining stability**.
- Understanding ecosystems is crucial for **environmental protection and sustainable development**.

Producers, Consumers, and Decomposers

Introduction

- All ecosystems work on the basis of **trophic levels**.
- Organisms are grouped into **producers, consumers, and decomposers**, based on how they obtain food.

Producers (Autotrophs)

1. Green plants, algae, some bacteria.
2. Capture **solar energy** → produce carbohydrates via photosynthesis.
3. Base of the **food chain**; support all higher organisms.

Consumers (Heterotrophs)

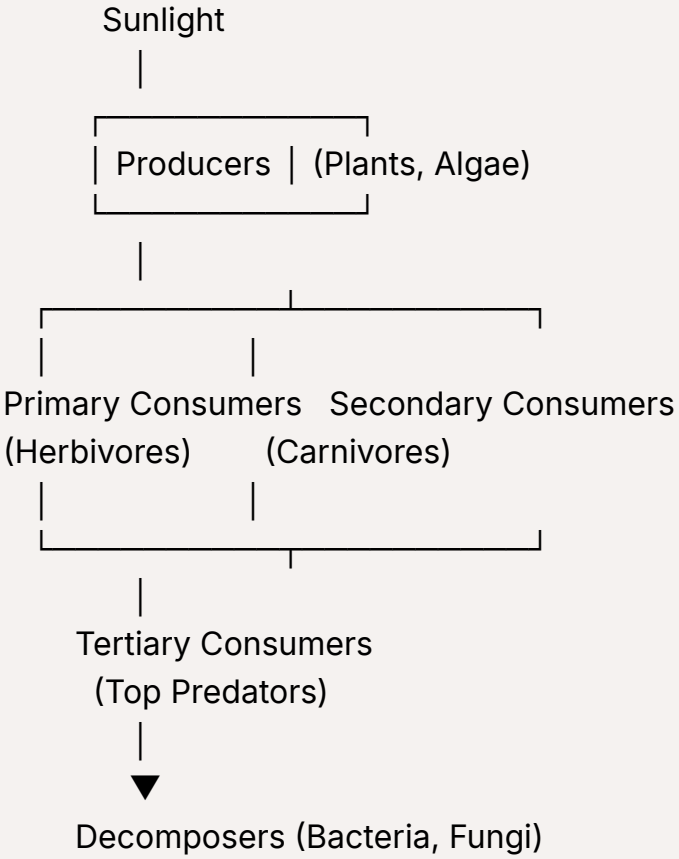
1. Depend on producers or other consumers.
2. Types:
    - **Primary consumers (herbivores):** Cow, deer, rabbit.
    - **Secondary consumers (small carnivores):** Frog, fox, snake.
    - **Tertiary consumers (top carnivores):** Lion, tiger, eagle.
    - **Omnivores:** Humans, bears.

Decomposers (Saprotrophs)

1. Bacteria, fungi, earthworms.

2. Break down dead organic matter → release nutrients back into soil.
3. Essential for **nutrient recycling** and soil fertility.

Flowchart



Conclusion

- Producers, consumers, and decomposers form the **foundation of ecosystem functioning**.
- Without any one group, **energy flow and nutrient cycling will collapse**.

Energy Flow in the Ecosystem

Introduction

- Energy flow is the **movement of energy through trophic levels** in an ecosystem.

- It is **unidirectional** (cannot be recycled).

Steps of Energy Flow

1. **Solar Energy Capture** → Plants trap sunlight via chlorophyll.
2. **Producers** → Convert light into chemical energy.
3. **Consumers** → Transfer energy by eating plants/animals.
4. **Decomposers** → Release nutrients, energy lost as heat.

Laws of Energy Flow

1. **First Law of Thermodynamics:** Energy is neither created nor destroyed, only transformed.
2. **Second Law of Thermodynamics:** Energy transfer is inefficient → some energy lost as heat.

10% Law (Lindeman's Law)

- Only **10% of energy** passes from one trophic level to the next.
- Example: Plants (1000J) → Herbivores (100J) → Carnivores (10J).

Flowchart

```
 Sunlight
    |
Producers (Plants)  →  Primary Consumers (Herbivores)
    |
Secondary Consumers (Carnivores)
    |
Tertiary Consumers (Top predators)
    |
Decomposers (Nutrient Recycling)
```

Conclusion

- Energy flow maintains **life processes and ecological balance**.

- As energy is lost at each level, ecosystems depend on **continuous solar input**.

Ecological Succession

Introduction

- Ecological succession = **gradual, natural process of change** in species composition in an area.
- Leads to a **stable climax community**.

Types of Succession

1. **Primary Succession** → On bare surfaces (rock, sand, volcanic lava). Starts from pioneer species (lichens, mosses).
2. **Secondary Succession** → On previously inhabited but disturbed land (forest fire, floods, agriculture fields).

Stages of Succession

1. **Pioneer stage** → First species, tolerate harsh conditions.
2. **Intermediate stage** → Grasses, shrubs, insects, small animals.
3. **Climax stage** → Stable community with trees, large animals.

Examples

- Rock → lichens → mosses → grasses → shrubs → forest.
- Pond → plankton → aquatic plants → reeds → forest.

Flowchart

```
Bare Land / Water
      |
Pioneer Stage (Lichens, Mosses)
      |
Intermediate Stage (Grasses, Shrubs, Insects)
      |
```

Climax Community (Forest, Stable Ecosystem)

Conclusion

- Succession restores **damaged ecosystems** and ensures **biodiversity and stability**.

Food Chains, Food Webs, and Ecological Pyramids

Food Chain

- **Definition:** A food chain is a linear sequence of organisms through which energy and nutrients flow, starting from producers to consumers and decomposers.
- **Example:** Grass → Grasshopper → Frog → Snake → Hawk
- **Types of Consumers in a Food Chain:**
  - **Primary consumers:** Herbivores (e.g., Grasshopper)
  - **Secondary consumers:** Carnivores feeding on herbivores (e.g., Frog)
  - **Tertiary consumers:** Carnivores feeding on other carnivores (e.g., Hawk)

Food Web

- **Definition:** A food web is a complex network of interconnected food chains in an ecosystem, showing the multiple feeding relationships.
- **Example:** Grass eaten by rabbits and deer; rabbits eaten by foxes and snakes; deer eaten by tigers.

Ecological Pyramids

- **Definition:** Graphical representation showing the **trophic levels** in an ecosystem.
- **Types:**
  1. **Pyramid of Numbers:** Number of organisms at each trophic level.
     - Example: 1000 grass → 100 grasshoppers → 10 frogs → 1 snake

2. **Pyramid of Biomass:** Total biomass at each trophic level.
    - Example: 1000 kg grass → 100 kg herbivores → 10 kg carnivores
3. **Pyramid of Energy:** Shows energy flow; always upright as energy decreases at each level.

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5. Ecosystems: Introduction, Types, Characteristics, Structure, and Functions

Introduction

- **Definition:** An **ecosystem** is a functional unit of nature where **living organisms (biotic) interact with non-living elements (abiotic)** in a particular area.
- Every ecosystem has **energy flow, nutrient cycling, and a dynamic equilibrium** between organisms and environment.
- **Examples:** Forest ecosystem, pond ecosystem, desert ecosystem.

Types of Ecosystems

1. **Terrestrial Ecosystems:** Land-based ecosystems
    - **Forests:** Tropical, temperate, coniferous
        - Example: Western Ghats (tropical rainforest)
    - **Grasslands:** Savannas, prairies
        - Example: Deccan plateau grasslands
    - **Deserts:** Low rainfall, sparse vegetation
        - Example: Thar Desert
2. **Aquatic Ecosystems:** Water-based ecosystems
    - **Freshwater:** Rivers, lakes, ponds
        - Example: River Ganga, Chilika Lake
    - **Marine:** Oceans, estuaries, coral reefs
        - Example: Arabian Sea, Andaman coral reefs

Characteristic Features

- **Energy Flow:** Sun → Producers → Consumers → Decomposers
- **Nutrient Cycling:** Carbon, nitrogen, and phosphorus are recycled
- **Self-sustaining:** Organisms and environment maintain balance
- **Dynamic:** Ecosystem changes due to succession or human activity

Structure

1. **Biotic Components:**
    - **Producers:** Plants, algae
    - **Consumers:** Herbivores, carnivores, omnivores
    - **Decomposers:** Fungi, bacteria
2. **Abiotic Components:**
    - Sunlight, temperature, water, soil, air

Functions

- **Energy flow:** Transfer of energy across trophic levels
- **Nutrient cycling:** Recycling of elements to maintain life
- **Habitat provision:** Shelter for organisms
- **Regulation of climate and gases:** CO_2 absorption by plants, oxygen production

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6. Biodiversity

Definition

- Biodiversity = **Variety of life on Earth**, including genes, species, and ecosystems.

Levels of Biodiversity

1. **Genetic Diversity:** Variation within species
    - Example: Different rice varieties, dog breeds
2. **Species Diversity:** Number of different species in an area
    - Example: Bengal tiger, Indian cobra, Mango tree

3. **Ecosystem Diversity:** Variety of habitats and ecological processes
    - Example: Mangroves, coral reefs, grasslands

Importance

- Ensures **ecosystem stability, food security, medicine, and ecological balance**

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7. Bio-geographical Classification of India

- India has **10 bio-geographical zones** classified based on climate, soil, and species composition:

Zone	Description	Example
1. Trans-Himalaya	Cold desert mountains	Ladakh
2. Himalaya	Alpine, temperate forests	Sikkim, Himachal
3. Indian Desert	Arid region with sparse vegetation	Thar Desert
4. Semi-arid	Moderate rainfall, scrub forests	Rajasthan plains
5. Western Ghats	High rainfall, evergreen forests	Kerala, Karnataka
6. Deccan Peninsula	Plateau with mixed vegetation	Maharashtra, Andhra Pradesh
7. Gangetic Plains	Fertile plains, riverine forests	Uttar Pradesh, Bihar
8. North-East India	Rainforests, high biodiversity	Assam, Meghalaya
9. Islands	Unique species, coastal ecosystems	Andaman, Nicobar, Lakshadweep
10. Coasts	Mangroves, estuaries	Sundarbans, Konkan Coast

Use: Helps in planning **conservation strategies and species protection**

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8. Value of Biodiversity

1. **Consumptive Use Value:** Direct use by humans
    - Food, timber, fuel, medicinal plants
    - Example: Fruits, honey, fish, bamboo
2. **Productive Use Value:** Commercial exploitation for trade

- Example: Agriculture (rice, wheat), pharmaceuticals

3. **Social Value:** Cultural and religious importance
   - Example: Sacred groves, temple ponds
4. **Ethical Value:** Moral duty to protect species
   - Example: Protecting elephants, tigers
5. **Aesthetic Value:** Beauty, recreation, tourism
   - Example: National parks, hill stations
6. **Option Value:** Potential future benefits
   - Example: Genes for crop improvement, new medicines

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9. Biodiversity at Global, National, and Local Levels

- **Global Level:** Earth has 8–9 million species; major hotspots in Amazon, Congo, Indonesia.
- **National Level (India):**
  - India has ~8% of the world's species despite only 2.4% of land area
  - Examples: Bengal tiger, Indian rhinoceros
- **Local Level:** Village forests, ponds, parks, gardens
  - Example: Urban parks, village ponds, community gardens

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10. India as a Mega-Diversity Nation

- **Definition:** Country with high species richness and endemism.
- **Reasons for mega-diversity:**
  - Varied climate zones (tropical, temperate, alpine)
  - Diverse ecosystems (forests, wetlands, deserts)
  - Geographic position in the tropics
- **Endemic Species:** Found only in India
  - Examples: Nilgiri tahr, Lion-tailed macaque

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11. Hotspots of Biodiversity

- **Definition:** Areas rich in species diversity and endemism but under threat.
- **Criteria:** At least 1500 species of vascular plants as endemics, lost 70% of original habitat.
- **India's 4 Biodiversity Hotspots:**
  1. **Himalaya** – High altitude, endemic plants
  2. **Indo-Burma** – Rich in freshwater species
  3. **Sundalands (Nicobar Islands)** – Unique island species
  4. **Western Ghats** – Tropical rainforests, endemic mammals and amphibians

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12. Threats to Biodiversity

1. **Habitat Loss:** Deforestation, urbanization, agriculture
2. **Poaching of Wildlife:** Illegal hunting for fur, tusks, horns
3. **Man–Wildlife Conflicts:** Crop damage by elephants, livestock predation
4. **Rare, Endangered, and Threatened (RET) Species:**
   - Endangered: Bengal tiger, Snow leopard
   - Vulnerable: Ganges river dolphin
   - Critically Endangered: Great Indian bustard
5. **Endemic Species:** Only found in India
   - Examples: Nilgiri langur, Indian giant squirrel

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13. Methods of Biodiversity Conservation

1. In-situ Conservation (in natural habitat)

- Protecting species **where they naturally live**
- Examples:
  - **National Parks:** Jim Corbett, Kaziranga
  - **Wildlife Sanctuaries:** Periyar, Ranthambore

- **Biosphere Reserves:** Nilgiri, Sundarbans

2. Ex-situ Conservation (outside natural habitat)

- Protecting species **outside their natural habitat**
- Examples:
  - **Zoos:** Safeguard animals like tigers, lions
  - **Botanical Gardens:** Preserve rare plants
  - **Seed Banks:** Store seeds for future use
  - **Cryopreservation:** Preserve germplasm or embryos